

HR WORKFORCE ANALYTICS DASHBOARD

SQL Server | Power BI

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Project Type: End-to-End HR Analytics Project

Tools Used: SQL Server Management Studio (SSMS), SQL, Power BI, DAX

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Executive Summary

Organizations rely on accurate workforce data to make informed decisions regarding recruitment, employee retention, and workforce planning. This project demonstrates the complete analytics workflow, beginning with a raw Human Resources dataset and ending with an interactive Power BI dashboard that provides meaningful business insights.

The dataset was cleaned, transformed, and analyzed using SQL Server before being imported into Power BI for visualization. The final dashboard enables HR professionals to monitor workforce demographics, hiring trends, employee retention, and individual employee records through interactive reports.

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1. Project Overview

The Human Resources department required a centralized reporting solution capable of transforming raw employee records into meaningful business intelligence. The original dataset contained inconsistencies, invalid values, duplicate records, and missing information that prevented accurate reporting.

This project demonstrates the complete lifecycle of a data analytics project, including:

- Data cleaning
- Data transformation
- SQL analysis
- KPI development
- Interactive dashboard design
- Business insight generation

The result is an HR analytics dashboard that supports informed workforce management and strategic decision-making.

2. Business Objectives

The primary objectives of this project were to:

- Clean and structure the HR dataset for analysis.
- Standardize employee information.
- Analyze workforce demographics.
- Monitor hiring trends over time.
- Evaluate employee retention.
- Measure workforce stability.
- Create an interactive dashboard for HR managers.
- Present actionable business insights.

3. Dataset Description

The dataset contains employee information across multiple organizational departments.

Key Fields Included

- Employee ID
- First Name
- Last Name
- Gender
- Race
- Department
- Job Title
- Hire Date
- Birth Date
- Termination Date
- City
- State

Additional calculated fields were created during the cleaning process, including:

- Employment Status
- Employee Age
- Years at Company

4. Data Cleaning & Preparation

Before analysis, the dataset underwent extensive cleaning using SQL Server.

Cleaning Activities Performed

Duplicate Detection

Duplicate employee records were identified and removed to ensure data integrity.

Date Standardization

Several date columns contained inconsistent formats.

The following fields were converted into SQL Date format:

- Hire Date
 - Birth Date
 - Termination Date
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Missing Values

Records containing incomplete information were reviewed and corrected where possible.

Employment Status

A calculated Employment Status field was created using the employee termination date.

Employees were classified as:

- Active
- Terminated

based on whether the termination date had passed.

Employee Age

Employee age was calculated using the difference between Birth Date and the current date.

Years at Company

Employee tenure was calculated from Hire Date to either:

- Today's date (Active Employees)
 - Termination Date (Former Employees)
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5. SQL Analysis

SQL queries were developed to answer key HR business questions.

Analyses Included

- Total employees
- Active employees
- Terminated employees
- Active rate
- Turnover rate
- Average employee tenure
- Employees by department
- Employees by gender
- Employees by race
- Employees by location
- Hiring trend by year
- Department retention
- Department turnover

Advanced SQL techniques included:

- CASE Statements
- Aggregate Functions
- CTEs (Common Table Expressions)
- Window Functions
- Ranking Functions

6. Dashboard Overview

The Power BI report consists of three interactive pages designed for different analytical purposes.

Dashboard	Purpose
Workforce Overview	Workforce demographics and organizational composition

Dashboard

Purpose

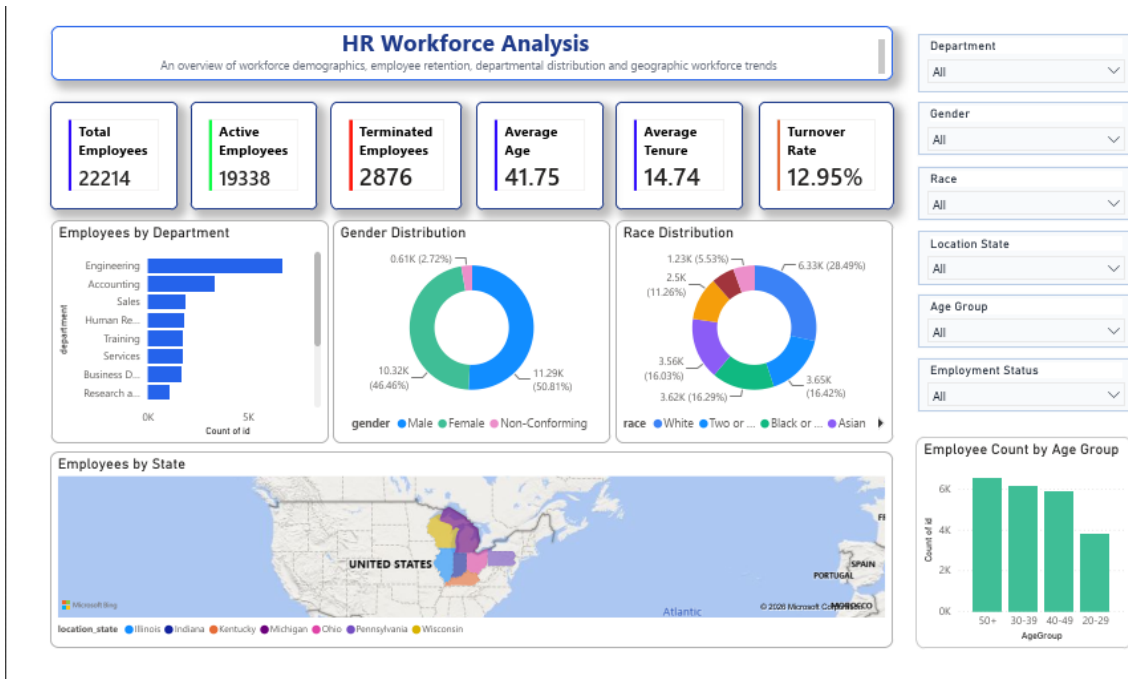
Hiring & Retention
Analysis

Hiring trends, employee retention, and workforce
stability

Employee Directory

Employee-level exploration and filtering

7. Workforce Overview



The Workforce Overview page provides a high-level summary of the organization's workforce composition.

KPIs Included

- Total Employees
- Active Employees
- Terminated Employees
- Average Age
- Average Tenure
- Turnover Rate

Visualizations

- Employees by Department
- Gender Distribution
- Race Distribution
- Employee Distribution by State
- Age Group Distribution

Purpose

This page helps HR managers understand:

- Workforce size
- Diversity
- Department distribution
- Geographic distribution
- Employee demographics

8. Hiring & Retention Analysis



This page focuses on workforce stability and recruitment patterns.

KPIs

- Total Employees
- Active Employees
- Average Tenure
- Active Rate
- Turnover Rate

Visualizations

- Employees Hired Per Year
- Active vs Terminated Employees
- Top 10 Departments by Average Tenure
- Top 8 Departments by Turnover Rate

Purpose

This page allows decision-makers to monitor:

- Hiring activity over time
- Employee retention
- Workforce stability
- Departmental performance

9. Employee Directory

Employee Directory

Search, filter and explore detailed employee records.

Department

All

Gender

All

Race

All

Employment Status

All

Location

All

Age Group

All

Employee ID	First Name	Last Name	Department	Job Title	Employment Status	Years At Company	Employee Age	gender	City	State
35-8348224	Lisetta	Broggini	Sales	Account Coordinator	Active	8	38	Male	Cleveland	Ohio
89-1495667	Fremont	Aguirre	Sales	Account Coordinator	Active	12	46	Male	Cleveland	Ohio
00-8527252	Leonor	Gernmet	Sales	Account Executive	Active	13	51	Male	Cleveland	Ohio
01-9817509	Garret	McDougle	Sales	Account Executive	Active	10	28	Female	Cleveland	Ohio
02-0702454	Shayne	Banford	Sales	Account Executive	Active	8	35	Female	Cleveland	Ohio
02-2892865	Amby	Nelius	Sales	Account Executive	Active	22	45	Male	Cleveland	Ohio
02-6180556	Aurea	Woolward	Sales	Account Executive	Active	26	51	Non-Conforming	Cleveland	Ohio
02-6313385	Meta	Dennerley	Sales	Account Executive	Terminated	1	51	Female	Cincinnati	Ohio
02-8962412	Agata	Bartholat	Sales	Account Executive	Active	17	53	Female	Chicago	Illinois
03-1185671	Farr	Le Fevre	Sales	Account Executive	Active	23	53	Female	Cleveland	Ohio
03-6346242	Somerset	De Freitas	Sales	Account Executive	Active	18	57	Male	Madison	Wisconsin
03-6456050	Temple	Camblin	Sales	Account Executive	Active	13	29	Male	Cleveland	Ohio
03-6968657	Alameda	Corben	Sales	Account Executive	Active	13	30	Male	Springfield	Illinois
03-8472914	Reagan	Meilhuish	Sales	Account Executive	Active	19	39	Male	Columbus	Ohio
03-8474248	Jocko	Kidder	Sales	Account Executive	Terminated	3	37	Male	Cleveland	Ohio
03-8848707	Juliet	Garford	Sales	Account Executive	Active	20	57	Female	Cleveland	Ohio
03-9110050	Wain	Cawley	Sales	Account Executive	Active	12	44	Male	Indianapolis	Indiana
03-9593488	Jillane	Webbe	Sales	Account Executive	Active	22	30	Female	Cleveland	Ohio
04-3494381	Ermin	Neachell	Sales	Account Executive	Active	8	42	Male	Cleveland	Ohio
04-4001420	Kordula	Enrique	Sales	Account Executive	Active	9	27	Female	Cleveland	Ohio
05-1408428	Thorny	Compford	Sales	Account Executive	Active	24	41	Female	Cleveland	Ohio
05-2946030	Halsey	Matteau	Sales	Account Executive	Active	16	27	Female	Cleveland	Ohio

The Employee Directory provides detailed employee records in a searchable format.

Filters Available

- Department
- Gender
- Race
- Employment Status
- Location
- Age Group

Purpose

The directory enables HR professionals to:

- Locate employees quickly
 - Filter workforce records
 - Explore employee details
 - Support operational HR tasks
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10. Key Business Insights

Workforce Overview

- The organization employs over 22,000 individuals.
 - 87.05% are currently active, while 12.95% have been terminated.
 - Employee distribution varies across departments, highlighting differences in workforce size.
 - Workforce diversity is represented across multiple demographic groups.
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Hiring Trends

- Hiring activity fluctuated across the years.
 - Recruitment peaked during certain periods before gradually stabilizing.
 - Recent hiring patterns indicate a more consistent recruitment strategy.
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Employee Retention

- Overall turnover rate is 12.95%, indicating a relatively stable workforce.
 - Average employee tenure of 14.74 years suggests strong long-term retention.
 - Turnover varies across departments, indicating that some areas experience greater employee exits than others.
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Department Performance

- Some departments demonstrate significantly higher average tenure, suggesting stronger employee retention.
 - Departments with higher turnover may benefit from additional workforce planning and retention initiatives.
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11. Business Recommendations

Based on the analysis, the following recommendations are proposed:

Improve Retention

Investigate departments with the highest turnover to identify potential causes such as workload, management practices, or career progression opportunities.

Strengthen Workforce Planning

Monitor hiring trends regularly to ensure recruitment aligns with organizational growth and staffing requirements.

Department Benchmarking

Use departments with high average tenure as benchmarks for retention strategies across the organization.

Ongoing Dashboard Monitoring

Maintain the Power BI dashboard as a live reporting solution to support evidence-based HR decision-making.

12. Skills Demonstrated

This project demonstrates proficiency in:

SQL

- Data Cleaning
- Data Transformation
- CASE Statements
- Aggregate Functions
- CTEs
- Window Functions
- Ranking Functions
- Business Reporting

Power BI

- Data Modeling
- DAX Measures
- KPI Development
- Interactive Dashboards
- Data Visualization
- Slicers
- Report Design

Data Analytics

- Data Preparation
- Exploratory Data Analysis
- Business Intelligence
- Dashboard Design
- Insight Generation
- Data Storytelling

13. Conclusion

This project demonstrates the complete end-to-end workflow of an HR analytics solution, beginning with raw employee data and culminating in an interactive Power BI dashboard.

By leveraging SQL Server for data preparation and Power BI for visualization, the project transforms raw HR records into meaningful business intelligence. The dashboard enables stakeholders to monitor workforce demographics, hiring trends, employee retention, and organizational performance through clear, interactive reporting.

The combination of data cleaning, analytical SQL queries, KPI development, and dashboard design showcases practical skills in data analytics and business intelligence while providing decision-makers with actionable insights to support workforce planning and strategic HR initiatives.
